Data Visualization - Milestone 2

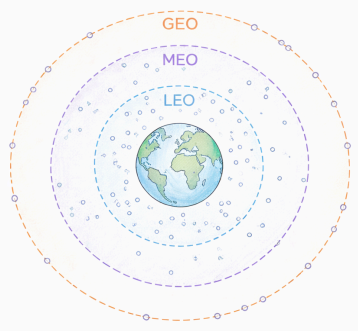
Team : 404-OrbitNotFound

1. Project Goal

The goal of this project is to help the users understand the distribution and usage of artificial satellites around Earth and time.

Instead of focusing on precise physical modeling, the project aims to provide an intuitive and accessible representation of satellite data, highlighting patterns related to ownership in different orbital regions, altitude distribution and in history.

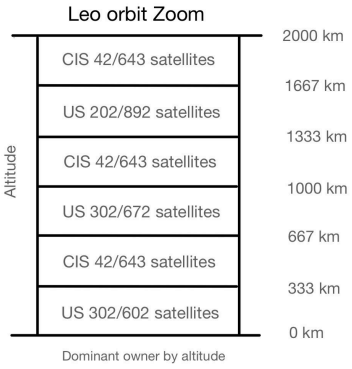
2. Visualization Sketches (Description)



There are three main visualisation:

2.1 Orbital Distribution of Satellites

- Three orbit zones (LEO, MEO, GEO) are shown as concentric rings around the earth
- Satellites are displayed as small points



A second visualization focuses on one of the orbits :

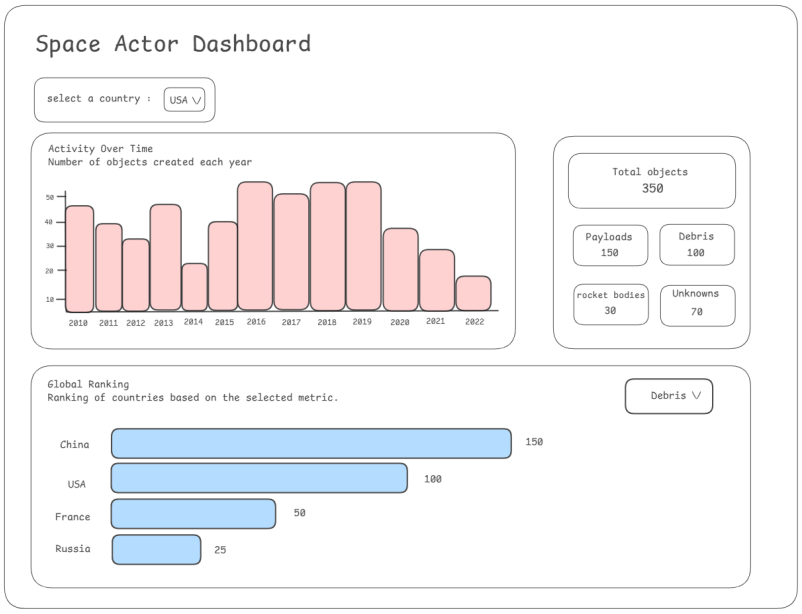
- The altitude range of the orbit is divided into a choosable number of segments
- Each segment shows the dominant country (owner) and the number of satellites

2.2 Look over your head

Based on the TLE provided by CelesTrak, we can obtain a large amount of information about satellites. This visualization consists of an interactive world map, and when a country is clicked, a list of satellites that have passed over it during a user-defined time period (from 1 hour to 48 hours) is displayed. When a satellite in the list is clicked, its trajectory over the same period, along with various details such as country of origin, launch site, launch date, speed, and altitude, appears.

2.3 Space Activity Dashboard by Country

The last visualization is an interactive dashboard focused on space actors. Its objective is to allow users to explore a country's space activity through various graphs. First, the user selects the country to analyze using a drop-down menu. A first graph then shows the number of objects launched per year. A second graph presents the composition of the objects, including the total number of objects, payloads, debris, rocket bodies, and unknowns. Finally, a last graph shows where the country ranks globally based on the different types of objects.



3. Tools

HTML, CSS, and JavaScript are used to build the structure and layout of the website, while SVG-based rendering (with a D3.js-inspired approach) is used to create the visualizations. Python libraries are used for data cleaning and early prototyping. Finally, GitHub and GitHub Pages are used for version control and deployment of the website.

4. Extra Features

Additional features could include linking the main visualization with the LEO zoom view, adding interactive elements such as hover or click to display satellite information, introducing animations or time-based evolution, and enabling comparisons between multiple countries. Improvements in styling, layout, and the use of color encoding or legends could also enhance readability. While these elements would enrich the user experience, they are not essential to the core functionality of the project. .

5. Functional Prototype

https://com-480-data-visualization.github.io/404-OrbitNotFound/